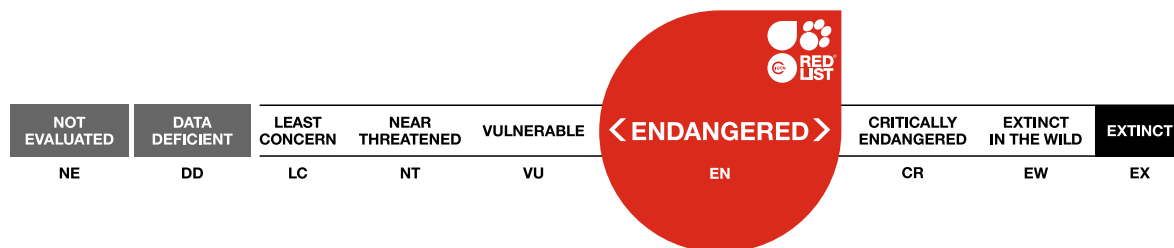


## *Sagittarius serpentarius*, Secretarybird

Assessment by: BirdLife International



View on [www.iucnredlist.org](http://www.iucnredlist.org)

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## Taxonomy

| Kingdom  | Phylum   | Class | Order           | Family        |
|----------|----------|-------|-----------------|---------------|
| Animalia | Chordata | Aves  | Accipitriformes | Sagittariidae |

**Scientific Name:** *Sagittarius serpentarius* (Miller, 1779)

**Common Name(s):**

- English: Secretarybird, Secretary Bird
- French: Serpentaire

**Taxonomic Source(s):**

del Hoyo, J., Collar, N.J., Christie, D.A., Elliott, A. and Fishpool, L.D.C. 2014. *HBW and BirdLife International Illustrated Checklist of the Birds of the World. Volume 1: Non-passerines*. Lynx Edicions BirdLife International, Barcelona, Spain and Cambridge, UK.

**Identification Information:**

A very large and distinctive terrestrial raptor, which stands c.1.2 m tall (Ferguson-Lees and Christie 2001). It is grey, whitish and black in all plumages, with small bill and head, bare face, relatively long neck, exceptionally long, bare legs, and long graduated tail with greatly elongated central rectrices. It has a distinctive crest of black-tipped spatulate feathers (Ferguson-Lees and Christie 2001).

## Assessment Information

**Red List Category & Criteria:** Endangered A2acde+3cde+4acde [ver 3.1](#)

**Year Published:** 2020

**Date Assessed:** April 30, 2020

**Justification:**

This species has been classified as Endangered because recent evidence from across its range suggests that its population is experiencing severe declines, probably owing to habitat degradation, disturbance, hunting and capture for trade.

**Previously Published Red List Assessments**

2016 – Vulnerable (VU)  
<https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T22696221A93549951.en>

2013 – Vulnerable (VU)  
<https://dx.doi.org/10.2305/IUCN.UK.2013-2.RLTS.T22696221A49946506.en>

2012 – Vulnerable (VU)

2011 – Vulnerable (VU)

2009 – Least Concern (LC)

2008 – Least Concern (LC)

2004 – Least Concern (LC)

2000 – Unknown (LR/LC)

1994 – Unknown (LR/LC)

1988 – Unknown (LR/LC)

## Geographic Range

### Range Description:

*Sagittarius serpentarius* occurs in sub-Saharan Africa, from southern **Mauritania, Senegal, The Gambia** and northern **Guinea** eastwards, north of the forest zone, through southern **Mali, Burkina Faso, Ghana, Togo, Benin**, southern areas of **Niger, Chad, Sudan, South Sudan** and northern parts of **Nigeria, Cameroon** and the **Central African Republic**, to **Ethiopia** and north-western **Somalia**, and south through eastern areas of the **Democratic Republic of Congo, Uganda, Kenya** and **Tanzania**, to southern Africa, including **Angola, Zambia, Malawi, Mozambique, Lesotho, Botswana, Zimbabwe, Namibia, Eswatini** and **South Africa**. It is variably described as common to rare (only 10 pairs or less in Waza-Logone west, Cameroon [J. Brouwer *in litt.* 2012]) and localised, and is sedentary in some parts of its range and nomadic in others (Ferguson-Lees and Christie 2001).

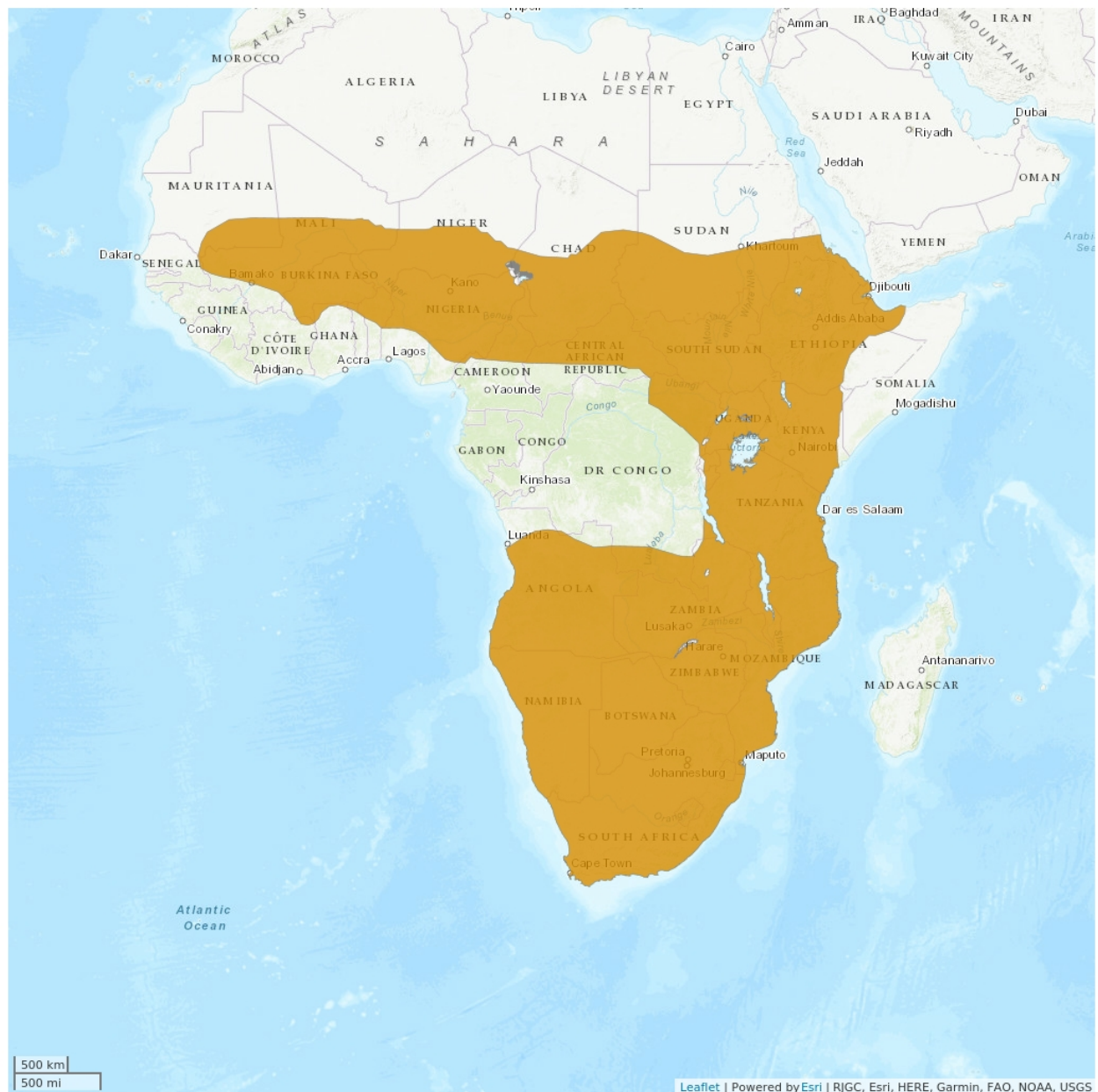
### Country Occurrence:

**Native, Extant (resident):** Angola; Benin; Botswana; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; Congo, The Democratic Republic of the; Côte d'Ivoire; Djibouti; Eritrea; Eswatini; Ethiopia; Gambia; Ghana; Kenya; Lesotho; Malawi; Mali; Mauritania; Mozambique; Namibia; Niger; Nigeria; Senegal; Somalia; South Africa; South Sudan; Sudan; Tanzania, United Republic of; Togo; Uganda; Zambia; Zimbabwe

**Extant & Vagrant:** Liberia; Rwanda

**Extant & Origin Uncertain:** Guinea-Bissau

# Distribution Map

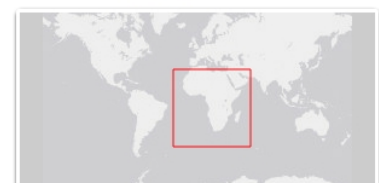


## Legend

EXTANT (RESIDENT)

## Compiled by:

BirdLife International and Handbook of the Birds of the World (2016) 2006



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

## Population

Although the species occurs across a vast range, surveyed densities suggest that the total population size does not exceed a five-figure number.

### Trend Justification

Large declines have been reported throughout the species's range, including the Sahel region of Mali, Niger, and Burkino Faso (Thiollay, 2001, R. Buij, unpublished data), Cameroon (Thiollay, 2006), Kenya (Ogada *et al.* 2020 *in prep.*), Botswana (Garbett *et al.*, 2018), Eswatini (Monadiem *et al.* 2003) and South Africa (Hofmeyr *et al.* 2014). On the basis of this evidence the species is inferred to be undergoing rapid declines overall.

**Current Population Trend:** Decreasing

## Habitat and Ecology (see Appendix for additional information)

The species inhabits open landscapes, ranging from open plains and grasslands, to lightly wooded savanna, but is also found in agricultural areas and sub-desert (Ferguson-Lees and Christie 2001), with up to 50% of recorded individuals in the Fynbos biome in winter being found in transformed environments (Hofmeyr *et al.* 2014). In Kruger National Park, South Africa, reports declined to zero in areas of >20% wood cover (Loftie-Eaton, 2017). It ranges from sea-level to 3,000 m. While it is nomadic, birds living in the moist grassland biome are less likely to be nomadic, but will travel on average 20-30 km per day while foraging (Kemp & Kemp, 1977; Whitecross *et al.* 2019).

Breeding occurs throughout the year and the species typically nests in a flat-topped *Acacia* or other thorny tree, where it constructs a flattened stick structure (Ferguson-Lees and Christie 2001). In Kenya, breeding attempts were observed to be disturbed by long droughts and unfavourable weather conditions (B. Amakobe, *in litt.* 2020). Juveniles can move a long way after leaving their nest site, but will return to their natal area (Retief and Smit-Robinson 2014; Whitecross *et al.* 2019). Juvenile males travel further than females, but returned closer to the nest site. Natal home ranges average  $1.21 \pm 0.34$  km<sup>2</sup> around the nest, and fledglings travel in an average  $2.62 \pm 0.37$  km straight-line distance from the nest (Whitecross *et al.* 2019). A 2 year old male was also recorded to have successfully raised two chicks in South Africa (Whitecross *et al.* 2017). A variety of prey is consumed, with insects forming 86% of the diet (Whitecross *et al.* 2019). Rodents, other mammals, lizards, snakes, eggs, young birds and amphibians also form part of the diet (Kemp and Kemp 1977; Ferguson-Lees and Christie 2001).

**Systems:** Terrestrial

## Use and Trade

Some reports of a few Secretarybirds being traded on traditional medicine markets, however the impact of this is believed to be low.

## Threats (see Appendix for additional information)

Although the species may benefit from deforestation, such positive effects may be outweighed by the negative impacts of spreading cultivation and urbanisation (Ferguson-Lees and Christie 2001). The excessive burning of grasslands may suppress populations of prey species, whilst the intensive grazing of livestock is also probably degrading otherwise suitable habitat (Baker *et al.* 2011). The Grassland Biome

in South Africa is threatened by the expansion of woody vegetation, which would translate to direct habitat loss and a possible reduction in foraging efficacy (Colyn *et al.*, 2020, *in review*). In Kenya, suitable habitat is being converted to other land uses, particularly for commercial purposes (B. Amakobe *in litt.* 2020). Disturbance by humans, probably most often herders, is likely to negatively affect breeding. The species is captured and traded in apparently small numbers; however, it is unknown how many die in captivity and transit. Direct hunting and nest-raiding for other uses and indiscriminate poisoning at waterholes are also potential threats. Exposure to secondary pesticide poisoning is a concern, and birds are also susceptible to negative impacts from collisions with fence lines and electric cables (Whitecross *et al.* 2019). 94 power-line fatalities have been recorded in 20 years by the Endangered Wildlife Trust (A. Botha *in litt.* 2020). Large ranging behaviour outside of protected areas increases their risk of injury and fatality (Whitecross *et al.* 2019). These human-induced threats may compound the effects of severe droughts in some areas (Baker *et al.* 2011).

## Conservation Actions (see Appendix for additional information)

### Conservation Actions Underway

CITES Appendix II. It occurs in a number of national parks and other protected areas across its large range. **Conservation Actions Proposed**

Initiate a coordinated continent-wide monitoring programme to obtain an up-to-date population estimate and track the species's trends. Engage with landowners of suitable properties to join biodiversity stewardship initiatives and to manage their properties in a sustainable way for Secretarybird populations. In areas where the species is declining, raise awareness of threats amongst local people, particularly livestock herders. Monitor and tackle the capture and trade of the species. Deploy more tracking devices to improve understanding of fine-scale movements of birds across the continent. Quantify current available, suitable habitat for the species across its range to assist with prioritisation of conservation and management of key sites.

## Credits

**Assessor(s):** BirdLife International

**Reviewer(s):** Wheatley, H.

**Contributor(s):** Amakobe, B., Baker, N., Botha, A., Brouwer, J., Buechley, E., Butchart, S., Colyn, R., Ekstrom, J., Howes-Whitecross, M., Ogada, D., Reteif, E., Shaw, P., Symes, A., Taylor, J. & Westrip, J.R.S.

**Facilitator(s) and Compiler(s):** Clark, J.

**Partner(s) and Institution(s):** BirdLife International

**Authority/Authorities:** IUCN SSC Bird Red List Authority (BirdLife International)

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## External Resources

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## Appendix

### Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                                                  | Season   | Suitability | Major Importance? |
|--------------------------------------------------------------------------|----------|-------------|-------------------|
| 2. Savanna -> 2.1. Savanna - Dry                                         | Resident | Suitable    | No                |
| 3. Shrubland -> 3.5. Shrubland - Subtropical/Tropical Dry                | Resident | Suitable    | No                |
| 4. Grassland -> 4.5. Grassland - Subtropical/Tropical Dry                | Resident | Suitable    | No                |
| 14. Artificial/Terrestrial -> 14.1. Artificial/Terrestrial - Arable Land | Resident | Suitable    | No                |
| 14. Artificial/Terrestrial -> 14.2. Artificial/Terrestrial - Pastureland | Resident | Suitable    | No                |

### Use and Trade

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| End Use                            | Local | National | International |
|------------------------------------|-------|----------|---------------|
| Medicine - human & veterinary      | No    | Yes      | No            |
| Pets/display animals, horticulture | No    | No       | Yes           |

### Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                      | Timing    | Scope                                                                                                     | Severity                   | Impact Score |
|-----------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------|----------------------------|--------------|
| 1. Residential & commercial development -> 1.1. Housing & urban areas                                                       | Ongoing   | Minority (50%)                                                                                            | Slow, significant declines | -            |
|                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation |                            |              |
| 2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.3. Agro-industry farming                    | Ongoing   | Minority (50%)                                                                                            | Slow, significant declines | -            |
|                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.1. Ecosystem conversion<br>1. Ecosystem stresses -> 1.2. Ecosystem degradation |                            |              |
| 2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.1. Nomadic grazing                                 | Ongoing   | Majority (50-90%)                                                                                         | Slow, significant declines | -            |
|                                                                                                                             | Stresses: | 1. Ecosystem stresses -> 1.2. Ecosystem degradation                                                       |                            |              |
| 5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target) | Ongoing   | Majority (50-90%)                                                                                         | Slow, significant declines | -            |
|                                                                                                                             | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                             |                            |              |

|                                                                                                                                       |           |                                                                                                         |                                  |   |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------|----------------------------------|---|
| 5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.2. Unintentional effects (species is not the target) | Ongoing   | Minority (50%)                                                                                          | Slow, significant declines       | - |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                           |                                  |   |
| 6. Human intrusions & disturbance -> 6.3. Work & other activities                                                                     | Ongoing   | Majority (50-90%)                                                                                       | Slow, significant declines       | - |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.2. Species disturbance<br>2. Species Stresses -> 2.3. Indirect species effects |                                  |   |
| 7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.1. Increase in fire frequency/intensity                        | Ongoing   | Minority (50%)                                                                                          | Slow, significant declines       | - |
|                                                                                                                                       | Stresses: | 1. Ecosystem stresses -> 1.2. Ecosystem degradation                                                     |                                  |   |
| 11. Climate change & severe weather -> 11.2. Droughts                                                                                 | Future    | Minority (50%)                                                                                          | Causing/could cause fluctuations | - |
|                                                                                                                                       | Stresses: | 2. Species Stresses -> 2.1. Species mortality                                                           |                                  |   |

## Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                               |
|---------------------------------------------------------------|
| <b>Conservation Action in Place</b>                           |
| In-place research and monitoring                              |
| Action Recovery Plan: No                                      |
| Systematic monitoring scheme: No                              |
| In-place land/water protection                                |
| Conservation sites identified: Yes, over entire range         |
| Occurs in at least one protected area: No                     |
| Invasive species control or prevention: No                    |
| In-place species management                                   |
| Successfully reintroduced or introduced benignly: No          |
| Subject to ex-situ conservation: No                           |
| In-place education                                            |
| Subject to recent education and awareness programmes: No      |
| Included in international legislation: No                     |
| Subject to any international management / trade controls: Yes |

## Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                                               |
|-------------------------------------------------------------------------------|
| <b>Conservation Action Needed</b>                                             |
| 3. Species management -> 3.1. Species management -> 3.1.1. Harvest management |
| 3. Species management -> 3.1. Species management -> 3.1.2. Trade management   |
| 4. Education & awareness -> 4.3. Awareness & communications                   |

## Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

|                                                            |
|------------------------------------------------------------|
| <b>Research Needed</b>                                     |
| 1. Research -> 1.2. Population size, distribution & trends |
| 3. Monitoring -> 3.1. Population trends                    |
| 3. Monitoring -> 3.3. Trade trends                         |

## Additional Data Fields

|                                                                       |
|-----------------------------------------------------------------------|
| <b>Distribution</b>                                                   |
| Continuing decline in area of occupancy (AOO): Unknown                |
| Extreme fluctuations in area of occupancy (AOO): No                   |
| Estimated extent of occurrence (EOO) (km <sup>2</sup> ): 23200000     |
| Continuing decline in extent of occurrence (EOO): Unknown             |
| Extreme fluctuations in extent of occurrence (EOO): No                |
| Continuing decline in number of locations: Unknown                    |
| Extreme fluctuations in the number of locations: No                   |
| <b>Population</b>                                                     |
| Number of mature individuals: 6,700-67,000                            |
| Continuing decline of mature individuals: Yes                         |
| Extreme fluctuations: No                                              |
| Population severely fragmented: No                                    |
| Continuing decline in subpopulations: Unknown                         |
| Extreme fluctuations in subpopulations: No                            |
| All individuals in one subpopulation: No                              |
| <b>Habitats and Ecology</b>                                           |
| Continuing decline in area, extent and/or quality of habitat: Unknown |
| Generation Length (years): 9.3                                        |

|                             |
|-----------------------------|
| <b>Habitats and Ecology</b> |
| Movement patterns: Nomadic  |

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